

RUIWEN WANG

PH.D. CANDIDATE | AI SYSTEMS | HPC | ML SYSTEMS

Sorbonne University & EURECOM | CIFRE with Huawei Paris Research Center

| Expected graduation: Aug 2026

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PROFILE

Ph.D. (CIFRE) candidate in Computer Science at Sorbonne University and EURECOM, in collaboration with Huawei Paris Research Center, working on efficient and reliable ultra-scale LLM and DNN training. I build symbolic performance and memory models, then couple them with ILP and constraint-based search to synthesize memory-feasible, communication-aware hybrid-parallel strategies and pipeline schedules. My work sits at the intersection of AI, HPC, and ML systems, has been validated on production Ascend clusters up to 16,384 NPUs, and has produced publications at CLUSTER, Euro-Par, NPC, and HPC Asia together with multiple patents and deployment-oriented planning workflows.

RESEARCH FOCUS

Doctoral thesis: *Systematic and Portable Optimisation of Hybrid Parallelism for Large-Scale Distributed Training* (Sorbonne University, doctoral school EDITE ED 130). A behaviour-level planning workflow built on a shared per-layer (Transformer) unit of reasoning, addressing one question: how can large-scale hybrid-parallel training be planned efficiently? Four capabilities, each realised as a system:

- **Predictability** — **PRISM**: a layer-level symbolic cost surface predicting compute, peak memory, and communication from model, training-config, and hardware descriptions, without profiling.
- **Interoperability** — **BMPipe + H2O**: factor planning into an inner schedule/memory problem and an outer choice of parallelism degrees (data, tensor, pipeline, virtual-pipeline, sequence, expert, optimiser), optimised jointly via ILP/MILP.
- **Evolvability** — **Concord**: a validity-preserving reuse contract that re-plans only the parts a model or configuration change invalidates, instead of restarting from scratch.
- **Adaptability** — **ManuMatic**: runtime strategy injection that corrects planning decisions when observed execution diverges from the abstraction used to choose them.

EXPERIENCE

Distributed and Parallel Computing Lab, Huawei Paris Research Center

Paris, France

Ph.D. Candidate (CIFRE) and Research Engineer – LLM Training Systems

2021 – 2026 (expected)

- Built cost-model-driven planners for hybrid and pipeline-parallel LLM training, covering stage partitioning, pipeline scheduling, and recomputation choices under memory and performance constraints.
- Designed symbolic peak-memory estimators and collective-cost models, then integrated overlap-aware strategies for large-scale throughput and MFU tuning.
- Translated prototypes into deployable planning workflows with engineering and customer teams, delivering configuration guidance, debuggable traces, and regression benchmarks across model families and scales.
- Supported technology transfer from research prototypes to production-scale training environments, including deployment feedback loops on customer workloads and long-sequence training scenarios.

Role progression: Research Apprentice (2021–2022) -> Research Engineer (2022–2023) -> CIFRE Ph.D. Candidate (2023–2026).

SELECTED IMPACT

- **BMPipe** (CLUSTER 2025): symbolic cost model plus ILP search over stage boundaries, intra-stage chunking, and per-layer recomputation; up to $1.36\times$ speedup with 83–92% HBM utilization, validated on clusters up to 16,384 Ascend NPUs.
- **ManuMatic** (NPC 2025): treated user-provided operator shardings as first-class planning constraints for robust “manual + automatic” search; $2.24\times$ on Mixtral-8 $\times$ 7B and $2.04\times$ on Llama3-8B vs. D-Rec, and +30.1% over an expert-tuned plan on Qwen2.5-72B.
- **H2O** (Euro-Par 2025, Best Poster): two-level workflow that searches parallelism degrees and micro-batch size, then refines layer-to-stage mapping and adaptive recomputation; MFU 26.13%→35.73% and +36.72% on DeepSeek-141B, transferred to production ($\approx 1.1\times$ on 4,096 devices; up to $2.37\times$ on long-sequence training at 512 devices).
- **PRISM** (HPC Asia 2026): profiling-free analytic planner predicting peak memory and communication; $1.23\times$ – $1.43\times$ speedups and MFU 29.40%→36.20%, validated up to 1,024 devices.

SELECTED
PUBLICATIONS

- **Ruiwen Wang**, Chong Li, Thibaut Tachon, Raja Appuswamy, Teng Su. *BMPipe: Bubble-Memory Co-optimization Strategy Planner for Very-Large DNN Training*. IEEE CLUSTER 2025.
- **Ruiwen Wang**, Chong Li, Raja Appuswamy, Yujie Yuan. *H2O: Holistic Hyperparameter Optimization for Large-Scale Deep Neural Network Training*. Euro-Par 2025. **Best Poster Award**.
- **Ruiwen Wang**, Chong Li, Hongxing Wang, Raja Appuswamy, Yujie Yuan. *ManuMatic: Strategy Injection for Robust Automatic Hybrid Parallelism in Distributed DNN Training*. NPC 2025.
- **Ruiwen Wang**, Philippe Fang, Chong Li, Thibaut Tachon, Raja Appuswamy. *PRISM: Profiling-Free Symbolic Memory-Driven Strategy Planner for Large DNN Model Training*. HPC Asia 2026.
- **Ruiwen Wang**, Thibaut Tachon, Chong Li, Raja Appuswamy. *Can Hybrid-Parallel Planning Support Alternating Model-Strategy Design? Dependency-Keyed Per-Layer Primitives for Re-Planning*. Under submission, 2026.
- Zhengdao Yu, **Ruiwen Wang**, Nelson Lossing, Chong Li. *MLLM Pipeline Bubble Modeling for Large-Scale Training*. HPC Asia 2026.
- Xinzhang Liu, Chao Wang, Zihua Yang, Zhuo Jiang, **Ruiwen Wang**, et al. *Training Report of TeleChat3-MoE*. arXiv:2512.24157, 2025.
- Chong Li, Pierre Leca, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *Devices and methods for generating execution plans for a machine learning model*. (WO 2025/020165 A1)
- Chong Li, **Ruiwen Wang**, Haoran Wang, Fan Yu. *A load balancing method to improve large model performance*.
- Chong Li, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *A multi-dimension parallelism configuration method for large language model training*.

PATENTS

EDUCATION

Sorbonne University & EURECOM	Paris, France
<i>Ph.D. in Computer Science</i>	2023 – 2026 (expected)
• Supervisors: Prof. Raja Appuswamy and Prof. Chong Li; doctoral school EDITE (ED 130) • Thesis: Systematic and Portable Optimisation of Hybrid Parallelism for Large-Scale Distributed Training	
Sorbonne University	Paris, France
<i>M.Sc. in Computer Science</i>	2019 – 2022
• Supervisors: Prof. Emmanuel Chailloux and Prof. Jacques Malenfant • Focus: Algorithms, Software Science, and Technology	
Paris-Saclay University	Paris, France
<i>B.Sc. in Computer Science</i>	2016 – 2019

RECOGNITION

Awards	Skills	
• Golden Prize, Huawei Challenge Award for Automatic Load Balancing 2024.08 • Best Poster Award, Euro-Par 2025.08 • Letter of Appreciation, China Telecom for DeepSeek-MoE training optimization deployment 2025.12	Programming: Python, C++, Java	Tooling: Ascend profiling, tracing, performance benchmarking
	ML stacks: MindSpore, PyTorch	Dev: Linux, Git
	Platforms: large-scale GPU/NPU clusters, cloud	Languages: Chinese, English, French, Cantonese